International and Monetary Economics

Course Code: 30054

Professor: Dmitriy Sergeyev

Lecture 1

Professor

Name: Dmitriy Sergeyev

Email: dmytro.sergeyev@unibocconi.it

Professor

Name: Dmitriy Sergeyev

Email: dmytro.sergeyev@unibocconi.it

MIPT
MOSCOW INSTITUTE OF PHYSICS AND TECHNOLOGY

NES
New Economic School

COLUMBIA UNIVERSITY
IN THE CITY OF NEW YORK

UNIVERSITÀ COMMERCIALE LUIGI BOCCONI MILANO

Roadmap

1. Course Outline
2. Course Logistics
3. Survey
4. (Maybe) Accounting

Course Title(s)

International and Monetary Economics

Alternatives

- Open-Economy Macro
- International Macro
- International Finance

Course Outline

1. Tools

2. Applications

Tools

1. Determinants of Exchange Rates
2. Determinants of Aggregate Output
3. Impact of Economic Policies

Some Applications

- History of the international monetary order
  - The Gold Standard, the Bretton Woods system, etc.
- Global imbalances
  - How do they look like? Where did they come from? Do they matter?
- Monetary Unions (e.g., the Euro Area)
  - Does it make sense to introduce a single currency? When?
- Macroeconomic policy spillovers
  - Should countries coordinate their policies?
- The Global Financial Cycle
  - What is it? What can explain it?
- The 2022-23 inflation surge
  - What happened? Why did it come down without a recession so far?

Course Logistics

Course Logistics

- **PDF** Syllabus (posted on blackboard) has lots of useful info (updated from time to time!)

- My office hours
  - TBA

30054 Instructors

Dmitriy Sergeyev (Classes 31-32)

Daniele Siena (Class 33)

Giulia Rivolta (Class 34)

30054 Teaching Assistants

Henri Graul (class 31)

Rouhan Song (Class 32)

Debmallya Chanda (Classes 33-34)

Grading

The final grade is based on

- 2 individual assignments
- 1 group project
- 2 in-class quizzes
- 1st partial, 2nd partial, general exams

Note: quizzes, individual and group assignment grades are valid for all <u>four</u> exam sessions (May, June, August, and January) even if previous attempts were failed

The Formula

$$\text{Exams grade} = \mathbb{I}\left(\text{Registered for the final exam}\right) \cdot \mathbb{I}\left(Final \geq 18\right) \cdot Final$$
$$+ \mathbb{I}\left(PE_1 \geq 18\right) \cdot \mathbb{I}\left(\text{Registered for the 2nd partial}\right) \cdot \frac{1}{2}\left[PE_1 + \mathbb{I}\left(PE_2 \geq 18\right) \cdot PE_2\right]$$

$$\text{Raw Grade} = 0.7 \cdot \text{Exams grade} +$$
$$+ 0.1 \cdot \max\left\{\text{Exams grade}, IA_1\right\} + 0.1 \cdot \max\left\{\text{Exams grade}, IA_2\right\}$$
$$+ 0.1 \cdot \max\left\{\text{Exams grade}, GA\right\} + \max\left\{Quiz_1, Quiz_2\right\}$$

$$\text{Final grade} = round\left(\text{Raw Grade}\right)$$

- *Final* – final exam (30 pts max)
- PE_1, PE_2 – partial exam (30 pts max)
- IA_1, IA_2, GA – individual assignments 1,2 and group assignments (30 pts max)
- $Quiz_1$, $Quiz_2$ – quizzes 1 and 2 scores (1 pnt max)
- $I(x)$ – an indicator function that equals 1 if X is true

Let’s unpack the formula

You can choose between **2 paths**.

<u>Option 1</u>

- *Exams grade* = partial exam 1 + partial exam 2
- **Conditions:**
  - **PE_1 (midterm)** must be ≥ 18
  - **PE_2 (second partial)** must also be ≥ 18
  - If both are valid, they contribute as their **average**

<u>Option 2</u>

- *Exams grade* = general exam

Raw grade

- **70%** of the grade comes from exams
- **30%** comes from:
  - individual assignments (IA_1, IA_2),
  - group assignment (GA),
  - quizzes
- The *max{Exams grade, assignment}* rule means assignments **can only help**, never lower the grade.
- + 1 extra point coming from Quizzes

Final grade

The final grade is simply the rounded raw grade.

Some caveats…

- The **second partial exam can be taken only once.**
- If a student:
  - registers for the **2nd partial** in the **first exam session**, they **cannot register again** for it in the second session
  - registers for the **general (final) exam** in the first session, they **lose the right** to take the 2nd partial in the second session
- What matters is **registration**, not attendance or passing!
- *Registering = using up the attempt*

$\Rightarrow$ Even if the student doesn't show up or fails, the attempt is consumed

Exercise 1

$$\begin{aligned}\text{Raw Grade} = 0.7 \cdot \text{Exams grade} &+ \\ &+ 0.1 \cdot \max\{\text{Exams grade}, IA_1\} + 0.1 \cdot \max\{\text{Exams grade}, IA_2\} \\ &+ 0.1 \cdot \max\{\text{Exams grade}, GA\} + \max\{Quiz_1, Quiz_2\}\end{aligned}$$

Luigi takes the general exam:

- $IA_1 = 15,\ IA_2 = GA = 0$
- $Exams\ grade\ = 30$
- $Quiz_1 = 0, Quiz_2 = 0.9$
- Raw Grade - ?
- Final Grade (= raw grade rounded) - ?

Exercise 2

$$\begin{aligned}\text{Exams grade} &= \mathbb{I}\left(\text{Registered for the final exam}\right)\cdot\mathbb{I}\left(Final \geq 18\right)\cdot Final \\ &\quad + \mathbb{I}\left(PE_1 \geq 18\right)\cdot\mathbb{I}\left(\text{Registered for the 2nd partial}\right)\cdot\frac{1}{2}\left[PE_1 + \mathbb{I}\left(PE_2 \geq 18\right)\cdot PE_2\right]\end{aligned}$$

$$\begin{aligned}\text{Raw Grade} &= 0.7\cdot\text{Exams grade}+ \\ &\quad + 0.1\cdot\max\left\{\text{Exams grade}, IA_1\right\} + 0.1\cdot\max\left\{\text{Exams grade}, IA_2\right\} \\ &\quad + 0.1\cdot\max\left\{\text{Exams grade}, GA\right\} + \max\left\{Quiz_1, Quiz_2\right\}\end{aligned}$$

Aurelia takes the two partial exams:

- $IA_1 = IA_2 = GA = 0$
- $PE_1 = 30, PE_2 = 29$
- $Quiz_1 = 1, Quiz_2 = 0$
- Raw Grade - ?
- Final Grade (=rounded raw grade) - ?

Exercise 3

$$\begin{aligned}\text{Raw Grade} = 0.7 \cdot \text{Exams grade} &+ \\ &+ 0.1 \cdot \max\{\text{Exams grade}, IA_1\} + 0.1 \cdot \max\{\text{Exams grade}, IA_2\} \\ &+ 0.1 \cdot \max\{\text{Exams grade}, GA\} + \max\{Quiz_1, Quiz_2\}\end{aligned}$$

Alexander takes the general exam

- $IA_1 = IA_2 = GA = 30$
- $Exams\ grade = 18$
- $Quiz_1 = 1, Quiz_2 = 1$
- Raw Grade - ?
- Final Grade (=rounded raw grade) - ?

Final Exam

- Similar structure to the exams of 2020-25 (see posted past exams on blackboard)
- Questions
  - multiple choice (essentially definitions)
  - a series of multiple-choice questions that constitute an analytic problem
  - TRUE/FALSE questions (with some explanations)
- Time
  - 60 minutes for those who take the second partial
- Score
  - 30 points

Quizzes

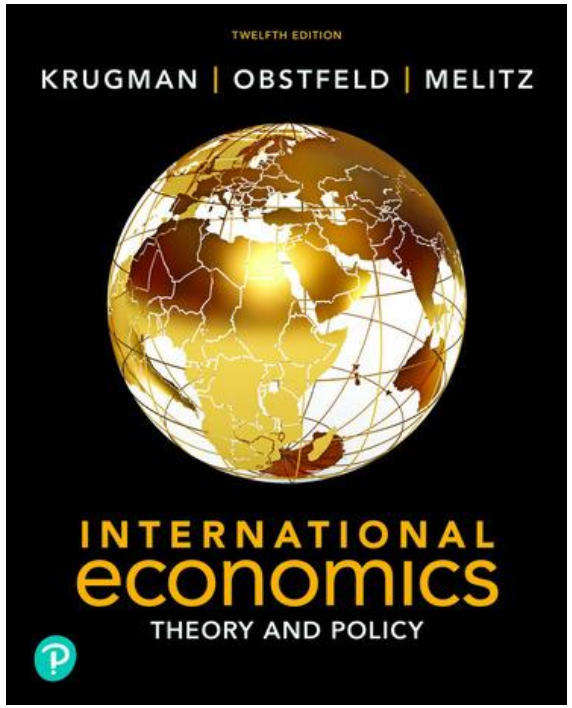
- Two graded quizzes
  - 10 multiple-choice questions
  - 10 minutes
  - Quiz 1 at the end of lecture 7 (March 2, 2026)
  - Quiz 2 at the end of lecture 19

Assignments

- 2 individual assignments
  - Gather and plot data
  - Solve analytic problems
  - Answer open-ended questions
  - Speculate
- 1 group (6 ppl max) assignment
  - An open-ended question related to our class
- Deadlines
  - IA1: March 19, 2026 at 23:59
  - IA2: May 14, 2026 at 23:59
  - GA: May 14, 2026 at 23:59

Material

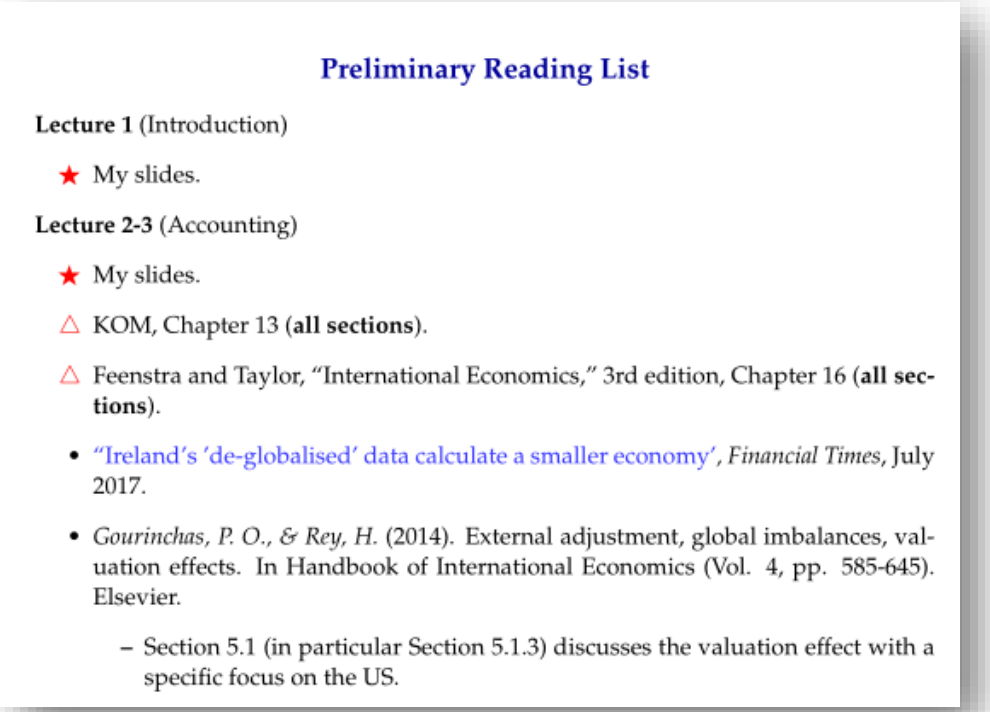
- Main Text
  - *International Economics: Theory and Policy (12e)* by Paul Krugman, Maurice Obstfeld, Marc Melitz
  - Should be available at the book store
  - Also available as e-book (cheaper)
  - 12th edition is not super different from the 11th edition



Material

- Slides (and other materials) are posted on
  - https://blackboard.unibocconi.it

- Only *starred* material in the syllabus is required for the exams



Preliminary Reading List

Lecture 1 (Introduction)

★ My slides.

Lecture 2-3 (Accounting)

★ My slides.

△ KOM, Chapter 13 (**all sections**).

△ Feenstra and Taylor, "International Economics," 3rd edition, Chapter 16 (**all sections**).

- "Ireland's 'de-globalised' data calculate a smaller economy', *Financial Times*, July 2017.
- *Gourinchas, P. O., & Rey, H.* (2014). External adjustment, global imbalances, valuation effects. In Handbook of International Economics (Vol. 4, pp. 585-645). Elsevier.
  - Section 5.1 (in particular Section 5.1.3) discusses the valuation effect with a specific focus on the US.

Message to Class Representatives

- **You have an extremely important and useful role!**

- Please introduce yourself by writing to me:
  dmytro.sergeyev@unibocconi.it

- Please find ways of adding students not from the BIEF program to your communication lines (WhatsApp, etc.)

Teams Group

- **Our course has a Teams Group**

- **Why to join?**
  - It is like a non-stop office-hour meeting: you can ask us questions, and everyone will see our answers
  - Combines all classes 31-34 in one space

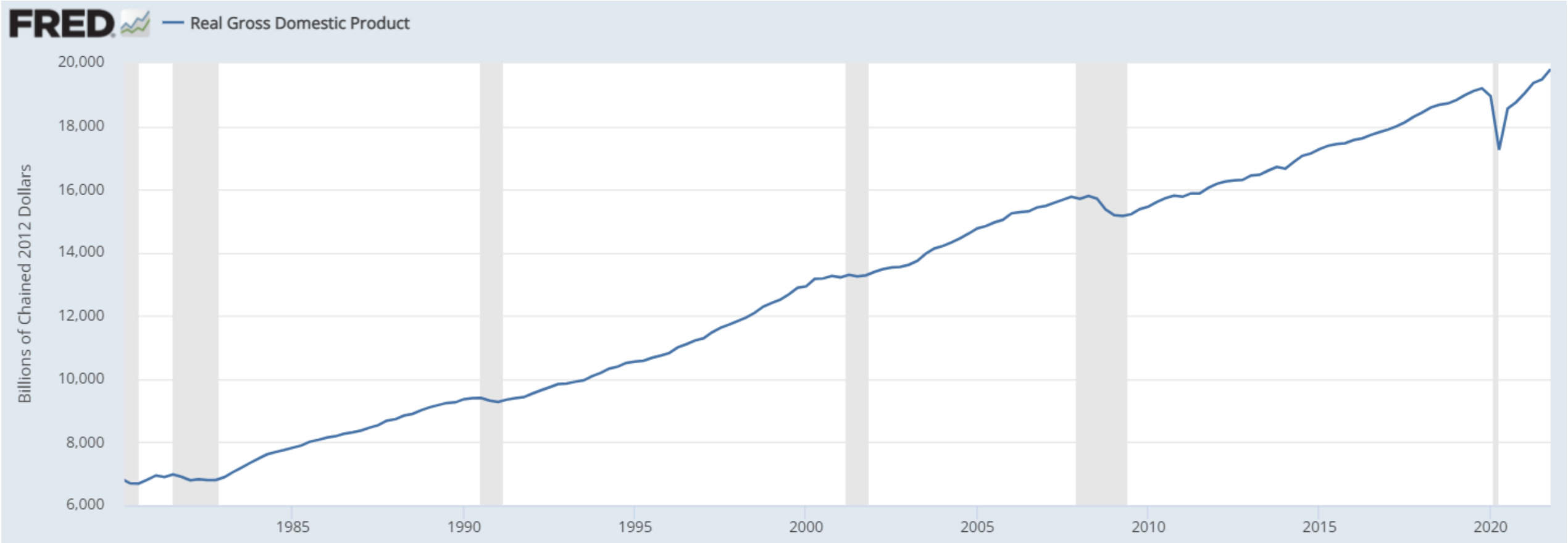
- **How to join?**
  - **On desktop:** Open Teams App -> Click Teams on the left vertical panel -> Click the “Join or create teams” button in the top right -> Insert the code **3m3eg4t** in the box “Join a team with a code”

Survey

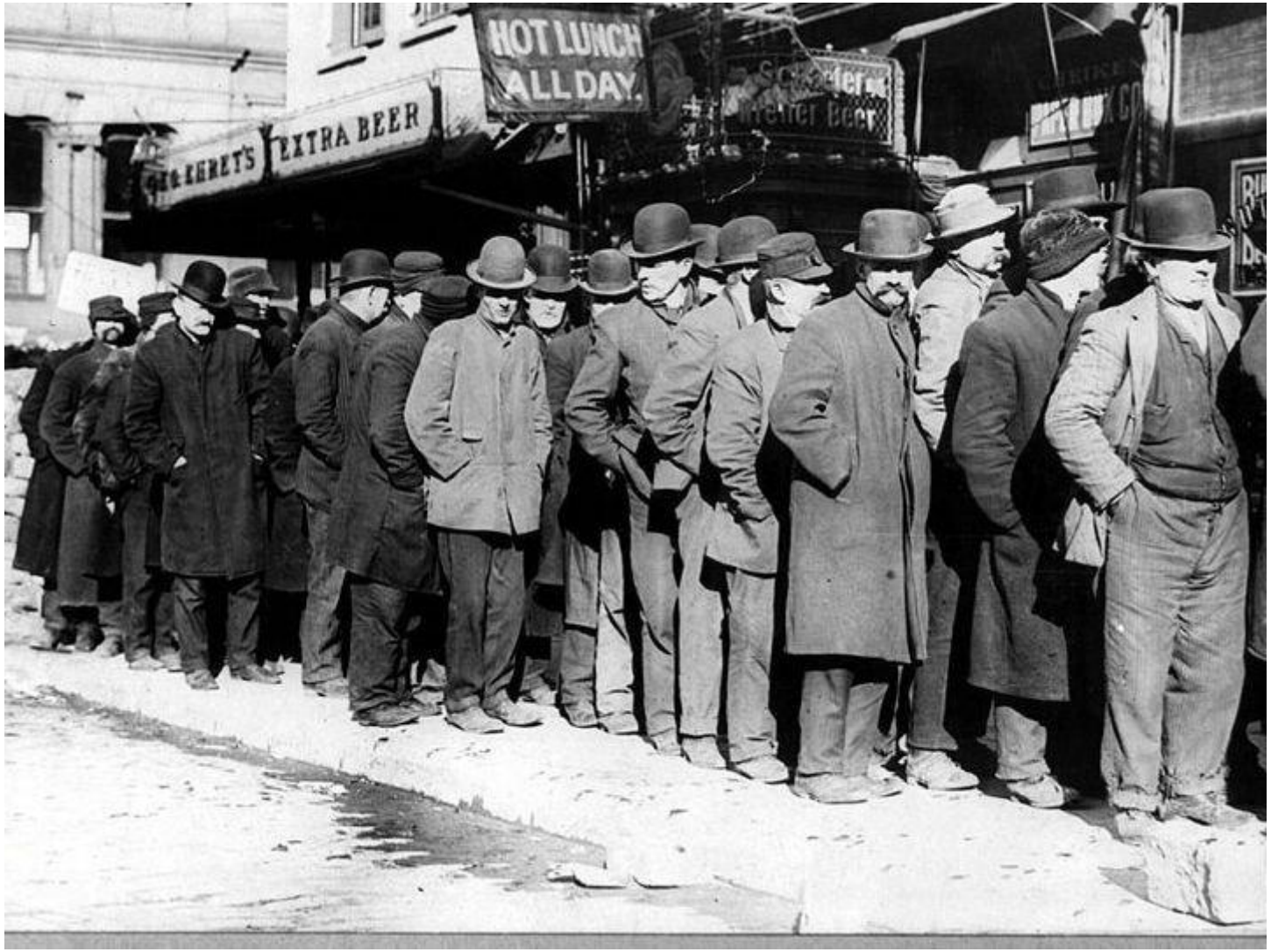
Type this link: https://tinyurl.com/228cnezb

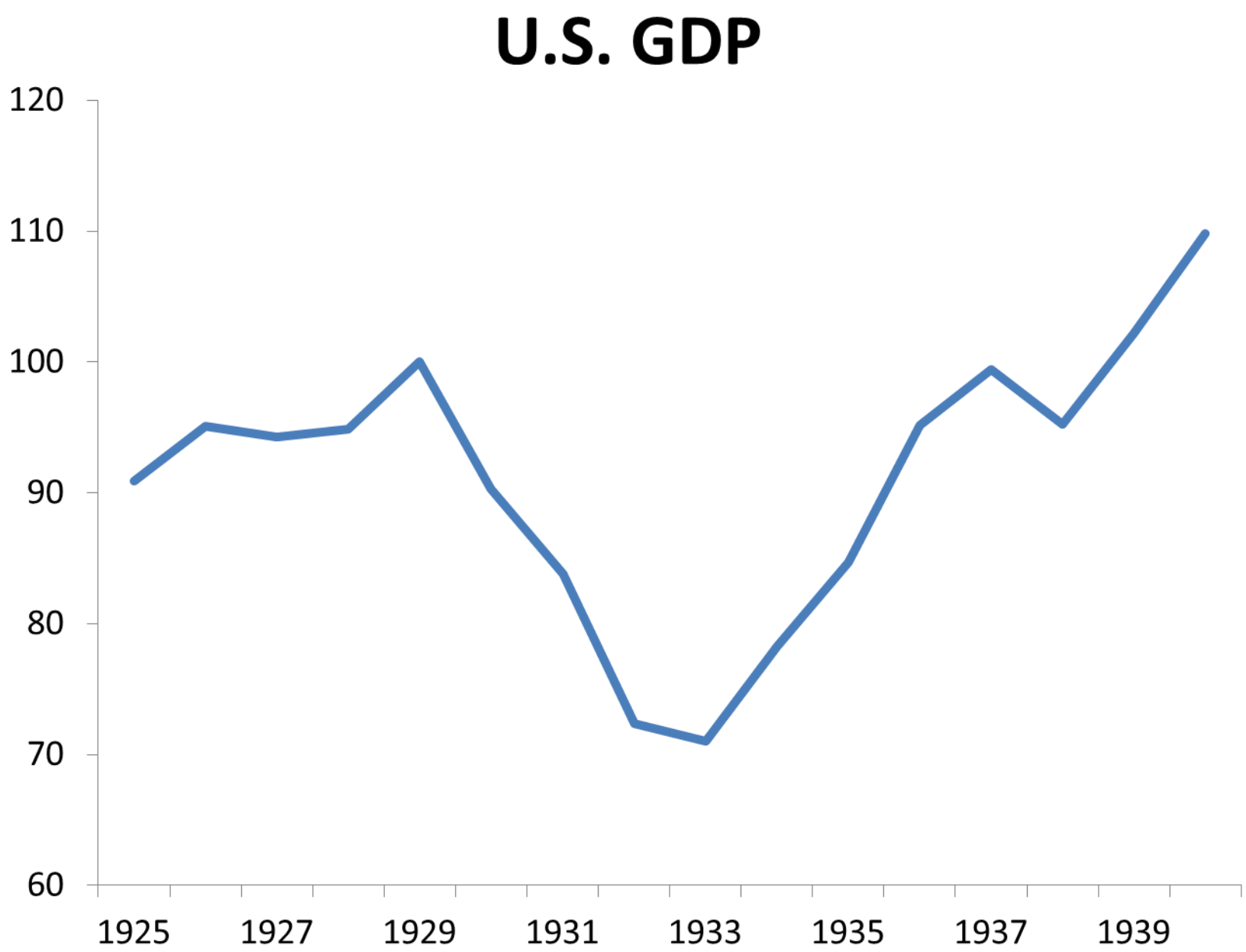
Or scan this code:

Accounting


FRED
Real Gross Domestic Product
Billions of Chained 2012 Dollars
20,000
18,000
16,000
14,000
12,000
10,000
8,000
6,000
1985
1990
1995
2000
2005
2010
2015
2020

The Great Depression




U.S. GDP
120
110
100
90
80
70
60
1925
1927
1929
1931
1933
1935
1937
1939

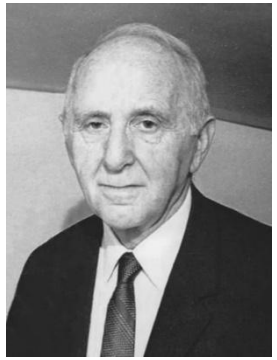
National Income and Product Accounts (NIPA)

- The Great Depression
  - Policymakers had only disaggregated info
    - stock prices, measures of industrial production, etc.
  - No measure of aggregate activity
    - To quantify the crisis
    - To assess effectiveness of policies
- Danger
  - rumors could cause self-fulfilling depression
- The recent financial crisis
  - Lack of information on banks interconnectedness

National Income and Product Accounts (NIPA)

- First constructed by Simon Kuznets and colleagues in 1930s
  - "One of the greatest inventions of the 20th century"

    William Nordhaus, Economist, Yale University, 2018 Nobel Prize



- Kuznets received Nobel Prize in 1971
- Richard Stone received Nobel Prize in 1984

Aggregate Economic Activity

An Overview

Gross Domestic Product

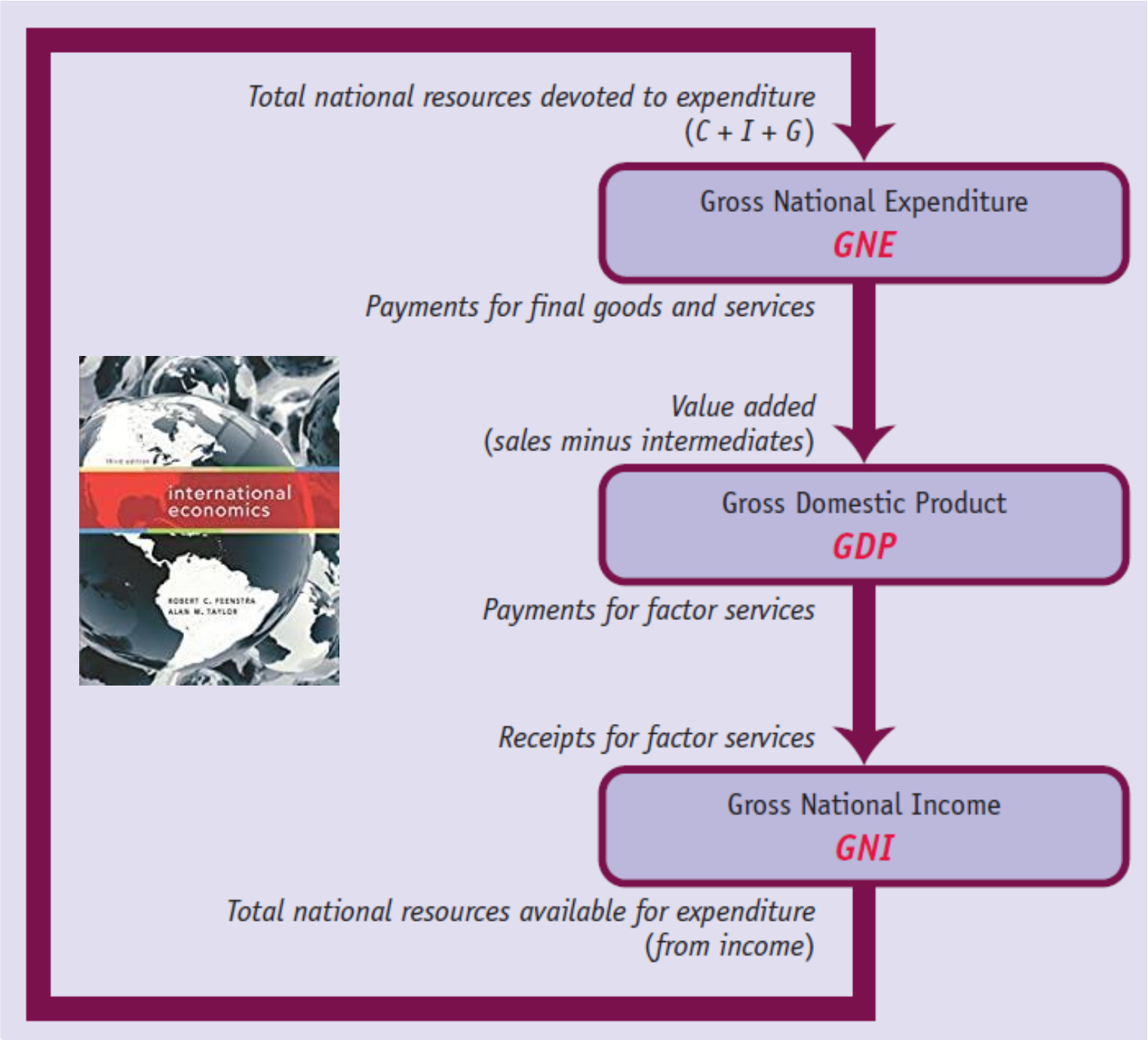
- **Gross domestic product** is the market value of the final goods and services newly produced in an economy over a certain period.

October 2025 Estimate for 2025 GDP

at the market exchange rate

- World: $117.2 trillion
- United States: $30.6 trillion
- European Union: $21.1 trillion
- China: $19.4 trillion
- Euro Area: $17.7 trillion
- United Kingdom: $4.0 trillion
- Italy: $2.5 trillion

Source: IMF



National Income and Product Accounts (**closed economy**)

- GNE – expenditure approach
- GDP – product approach
- GNI – income approach

In a closed economy

- **GNE=GDP=GNI**

Source: International Economics, Chapter 16, Feenstra and Taylor, 3rd edition

Example of the 3 approaches

Luigi grows olives on his land, hires some workers to harvest them, and sells olives at a market place.

- Expenditure approach
  - Count purchased olives
- Product approach
  - Count Luigi's olives
- Income approach
  - Workers' wages + Luigi's "profit" (return to Luigi's labor)

Recall what is counted as investment

- Business fixed investment (nonresidential)
  - spending by firms on plants, machinery, and equipment
- Residential investment
  - construction of new houses and apartment buildings
- Inventory investment
  - changes in inventories (of final or intermediate goods)

Recall Product Approach

- Idea
  - add up only the value of final goods
- Issue with intermediate goods
  - Do not double-count, instead use value-added
- Used goods
  - Do not count
- Inventories (= new unsold goods)
  - If they do not perish, count as investment

Next Time

- Next lecture:
  - KOM, Chapter 13